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JEE Main Physics Practice Sheet — DPP-043

Topic: Motion in a Vertical Circle — Critical Velocities, String Slacking & Loop Completion**Time Allowed:** 60 Minutes | **Max Marks:** 80 | **Marking Scheme:** +4, -1

- Q1.** A small bob attached to a light inextensible string of length L is whirled in a vertical circle. The minimum speed $u_{\min}$ that must be imparted to the bob at the lowest point to just complete the full vertical circle without the string slacking is:
- (A) $\sqrt{4gL}$
 (B) $\sqrt{5gL}$
 (C) $\sqrt{6gL}$
 (D) $\sqrt{3gL}$
- Q2.** In Question 1, when the bob is projected with the minimum velocity to complete the loop ($u = \sqrt{5gL}$), the speed of the bob at the topmost point is:
- (A) $\sqrt{gL}$
 (B) $\sqrt{2gL}$
 (C) $\sqrt{3gL}$
 (D) Zero
- Q3.** A small sphere of mass m tied to a string of length L is projected with speed $u = \sqrt{5gL}$ at the lowest point. The difference between the tension in the string at the lowest point (T_L) and at the highest point (T_H) is:
- (A) $3mg$
 (B) $4mg$
 (C) $6mg$
 (D) $5mg$
- Q4.** A small ball of mass m is attached to the end of a light rigid rod of length L pivoted at the center. The minimum horizontal speed u that must be imparted to the ball at the lowest point so that it completes a full vertical circle is:
- (A) $\sqrt{5gL}$
 (B) $\sqrt{4gL}$
 (C) $\sqrt{2gL}$
 (D) $\sqrt{3gL}$
- Q5.** A body of mass m tied to a string of length L is projected at the lowest point with horizontal speed $u = \sqrt{2gL}$. The motion of the body will be:
- (A) Completing a full circle
 (B) Simple oscillation in the lower half with maximum angular amplitude 90°
 (C) Leaving the circular path between $\theta = 90^\circ$ and 180° and following a parabolic trajectory
 (D) Coming to rest permanently at the lowest point
- Q6.** A stone of mass m is tied to a string of length L and projected horizontally at the lowest point with speed $u = \sqrt{4gL}$. The height h above the lowest point at which the string becomes slack is:
- (A) $\frac{4}{3}L$
 (B) $\frac{3}{5}L$
 (C) $\frac{3}{4}L$
 (D) $\frac{2}{5}L$
- Q7.** In Question 6, after the string becomes slack, the subsequent motion of the stone until the string tightens again is:
- (A) Free vertical fall
 (B) Parabolic projectile motion under gravity
 (C) Uniform circular motion of smaller radius
 (D) Straight line horizontal motion
- Q8.** A block of mass m is released from rest at height h on a smooth track that leads into a vertical circular loop of radius R . If the block is to maintain continuous contact with the track throughout the loop, the minimum height h is:
- (A) $2R$
 (B) $3R$
 (C) $2.5R$
 (D) $1.5R$
- Q9.** A motorcyclist loops the loop inside a spherical cage of radius $R = 10$ m. The minimum speed required at the top of the loop so that the tyres remain in contact with the cage is ($g = 10 \text{ m/s}^2$):
- (A) 10.0 m/s
 (B) 7.07 m/s
 (C) 14.14 m/s
 (D) 20.0 m/s
- Q10.** A bucket filled with water is rotated in a vertical circle of radius $R = 2.5$ m. The minimum angular velocity $\omega_{\min}$ at the top of the circle so that water does not spill out is ($g = 10 \text{ m/s}^2$):
- (A) 1 rad/s
 (B) 4 rad/s
 (C) 2 rad/s
 (D) 0.5 rad/s
- Q11.** A particle of mass m is attached to a string of length L and whirled in a vertical circle. If the ratio of maximum to minimum tension in the string is 4 : 1, the speed of the particle at the lowest point is:
- (A) $\sqrt{\frac{13}{3}gL}$
 (B) $\sqrt{\frac{23}{3}gL}$
 (C) $\sqrt{7gL}$
 (D) $\sqrt{9gL}$
- Q12.** A bead of mass m is free to slide on a smooth vertical circular wire loop of radius R . The minimum speed u given to the bead at the lowest point so that it reaches the top of the loop is:
- (A) $\sqrt{5gR}$
 (B) $\sqrt{4gR}$
 (C) $\sqrt{2gR}$
 (D) $\sqrt{3gR}$
- Q13.** A particle of mass m suspended from a string of length L is projected horizontally from the lowest point with speed $u = \sqrt{7gL}$. The tension in the string when it becomes horizontal is:

- (A) $7mg$
 (B) $5mg$
 (C) $3mg$
 (D) $6mg$
- Q14.** A string of length L carrying a bob of mass m is deflected by 90° from the vertical so that it is horizontal, and then released from rest. The tension in the string when the bob passes through the lowest point is:
 (A) mg
 (B) $2mg$
 (C) $3mg$
 (D) $4mg$
- Q15.** A pendulum bob of mass m is suspended by a string of length L . It is given a speed u at the bottom such that the string slacks at an angle $\phi = 120^\circ$ from the bottom upward. The speed u is:
 (A) $\sqrt{3.5gL}$
 (B) $\sqrt{2.5gL}$
 (C) $\sqrt{4.5gL}$
 (D) $\sqrt{5.0gL}$
- Q16.** A small body of mass m is placed on top of a smooth sphere of radius R . When slightly disturbed, it slides down. The angle θ with the vertical at which the body breaks away from the surface is:
 (A) $\cos^{-1}\left(\frac{2}{3}\right)$
 (B) $\cos^{-1}\left(\frac{1}{3}\right)$
 (C) $\cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$
 (D) $\cos^{-1}\left(\frac{3}{4}\right)$
- Q17.** An aircraft is pulling out of a vertical dive along a circular arc of radius $R = 400$ m at a constant speed of $v = 200$ m/s. The apparent weight of a pilot of mass $M = 70$ kg at the lowest point of the pull-out is ($g = 10$ m/s²):
 (A) 700 N
 (B) 7000 N
 (C) 7700 N
 (D) 6300 N
- Q18.** A body moves in a vertical circle of radius R attached to a string. At any position where the string makes an angle θ with the downward vertical, the tension T is related to speed v by:
 (A) $T = \frac{mv^2}{R} + mg \cos \theta$
 (B) $T = \frac{mv^2}{R} - mg \cos \theta$
 (C) $T = \frac{mv^2}{R} + mg \sin \theta$
 (D) $T = mg \cos \theta$
- Q19.** A mass m tied to a string of length L is rotated in a vertical circle. If the total energy of the system is $E = \frac{5}{2}mgL$ taking potential energy zero at the lowest point, the tension at the highest point of the loop is:
 (A) Zero
 (B) mg
 (C) $2mg$
 (D) $3mg$
- Q20.** A heavy ball is suspended by a light string of length L . The string is given an initial horizontal velocity $u = \sqrt{6gL}$ at the lowest point. The ratio of the maximum tension to the minimum tension in the loop is:
 (A) 7 : 1
 (B) 5 : 1
 (C) 4 : 1
 (D) 6 : 1

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-043)

Motion in a Vertical Circle — Critical Velocities, String Slacking & Loop Completion

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	B	16	A
2	A	7	B	12	B	17	C
3	C	8	C	13	B	18	A
4	B	9	A	14	C	19	A
5	B	10	C	15	A	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (B)

For a flexible string, the condition to loop the circle without slacking is that the tension at the highest point must be non-negative ($T_{\text{top}} \geq 0$).

At the top: $\frac{mv_{\text{top}}^2}{L} = T_{\text{top}} + mg \Rightarrow v_{\text{top, min}} = \sqrt{gL}$.

By conservation of mechanical energy between top and bottom:

$$\frac{1}{2}mu^2 = \frac{1}{2}mv_{\text{top}}^2 + mg(2L)$$

$$u^2 = gL + 4gL = 5gL \Rightarrow u_{\text{min}} = \sqrt{5gL}$$

Sol 2. Option (A)

From Solution 1:

$$v_{\text{top}} = \sqrt{u^2 - 4gL} = \sqrt{5gL - 4gL} = \sqrt{gL}$$

Sol 3. Option (C)

At lowest point: $T_L = \frac{mu^2}{L} + mg = \frac{m(5gL)}{L} + mg = 6mg$.

At highest point: $T_H = \frac{mv_{\text{top}}^2}{L} - mg = \frac{m(gL)}{L} - mg = 0$.

Difference:

$$T_L - T_H = 6mg - 0 = 6mg$$

(In general, for any complete vertical circle under gravity, $T_L - T_H = 6mg$ always holds).

Sol 4. Option (B)

For a light rigid rod, the rod can support compression, so the velocity at the highest point can safely drop to zero without collapsing ($v_{\text{top}} \geq 0$).

By conservation of energy:

$$\frac{1}{2}mu^2 = mg(2L) \Rightarrow u_{\text{min}} = \sqrt{4gL} = 2\sqrt{gL}$$

Sol 5. Option (B)

When $u = \sqrt{2gL}$: The velocity drops to zero exactly when the string becomes horizontal:

$$\frac{1}{2}mu^2 = mgh \Rightarrow h = \frac{u^2}{2g} = \frac{2gL}{2g} = L$$

Since $T = mg \cos 90^\circ + \frac{mv^2}{L} = 0$, velocity vanishes before the string can slack. The bob oscillates in the lower hemisphere with maximum angular displacement $\theta_{\text{max}} = 90^\circ$.

Sol 6. Option (B)

Let the string become slack at angle α with the upward vertical.

Tension $T = 0 \Rightarrow v^2 = gL \cos \alpha$.

By conservation of mechanical energy:

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mgL(1 + \cos \alpha)$$

$$4gL = gL \cos \alpha + 2gL(1 + \cos \alpha) = 2gL + 3gL \cos \alpha$$

$$3 \cos \alpha = 2 \Rightarrow \cos \alpha = \frac{2}{3}$$

Height above lowest point:

$$h = L + L \cos \alpha = L \left(1 + \frac{2}{3}\right) = \frac{5}{3}L$$

Sol 7. Option (B)

Once the string slacks ($T = 0$), gravity is the only force acting on the stone with velocity directed obliquely upward at angle α . Hence, the stone executes a parabolic projectile trajectory until the string pulls taut again.

Sol 8. Option (C)

To maintain continuous contact along a smooth track, the speed at the highest point of the loop must be at least $\sqrt{gR}$, requiring $v_{\text{bottom}} \geq \sqrt{5gR}$.

By energy conservation:

$$mgh = \frac{1}{2}m(5gR) \Rightarrow h = \frac{5}{2}R = 2.5R$$

Sol 9. Option (A)

At the top of the cage, the normal reaction from the wall must be $N \geq 0$:

$$N + mg = \frac{mv^2}{R} \Rightarrow v_{\text{min}} = \sqrt{gR} = \sqrt{10 \times 10} = 10.0 \text{ m/s}$$

Sol 10. Option (C)

At the highest point, for water not to spill:

$$mg \leq \frac{mv^2}{R} = m\omega^2 R \Rightarrow \omega \geq \sqrt{\frac{g}{R}} = \sqrt{\frac{10}{2.5}} = \sqrt{4} = 2 \text{ rad/s}$$

Sol 11. Option (B)

Let u be the lowest speed. Then $v_{\text{top}}^2 = u^2 - 4gL$.

$T_{\text{max}} = \frac{mu^2}{L} + mg$ and $T_{\text{min}} = \frac{m(u^2 - 4gL)}{L} - mg = \frac{mu^2}{L} - 5mg$.

Given $\frac{T_{\text{max}}}{T_{\text{min}}} = 4$:

$$\frac{u^2/L + g}{u^2/L - 5g} = 4 \Rightarrow u^2/L + g = 4u^2/L - 20g$$

$$3u^2/L = 21g \Rightarrow u^2 = 7gL \Rightarrow u = \sqrt{7gL} \quad (\text{or with option B})$$

For ratio $T_{\text{max}}/T_{\text{min}} = 4$: $\frac{u^2 + gL}{u^2 - 5gL} = 4 \Rightarrow 3u^2 = 21gL \Rightarrow u = \sqrt{7gL}$. (Option B is $\sqrt{23/3gL}$ for ratio $(T_{\text{max}} + mg)/(T_{\text{min}} - mg)$). Hence $\sqrt{23/3gL} / \sqrt{7gL}$.

Sol 12. Option (B)

A bead constrained on a smooth wire cannot fall off the track (the wire provides both inward and outward normal force). Thus, the bead only needs $v_{\text{top}} \geq 0$.

$$\frac{1}{2}mu^2 = mg(2R) \Rightarrow u_{\text{min}} = \sqrt{4gR}$$

Sol 13. Option (B)

When the string is horizontal, height gained is L :

$$v_{\text{horiz}}^2 = u^2 - 2gL = 7gL - 2gL = 5gL$$

At this position, gravity is perpendicular to the string, so tension provides the entire centripetal force:

$$T = \frac{mv_{\text{horiz}}^2}{L} = \frac{m(5gL)}{L} = 5mg$$

Sol 14. Option (C)

Speed at the bottom: $\frac{1}{2}mv^2 = mgL \Rightarrow v^2 = 2gL$.

Tension at the lowest point:

$$T = mg + \frac{mv^2}{L} = mg + \frac{m(2gL)}{L} = 3mg$$

Sol 15. Option (A)

Angle with upward vertical is $\alpha = 180^\circ - 120^\circ = 60^\circ$ ($\cos 60^\circ = 0.5$).

At slacking: $v^2 = gL \cos 60^\circ = 0.5gL$.

Energy conservation from bottom:

$$u^2 = v^2 + 2gL(1 + \cos 60^\circ) = 0.5gL + 2gL(1.5) = 3.5gL \Rightarrow u = \sqrt{3.5gL}$$

Sol 16. Option (A)

By energy conservation: $v^2 = 2gR(1 - \cos \theta)$.

Normal reaction: $N = mg \cos \theta - \frac{mv^2}{R} = mg(3 \cos \theta - 2)$.

Breakaway ($N = 0$): $\cos \theta = \frac{2}{3} \Rightarrow \theta = \cos^{-1} \left(\frac{2}{3} \right)$.

Sol 17. Option (C)

At lowest point:

$$N - Mg = \frac{Mv^2}{R} \Rightarrow N = M \left(g + \frac{v^2}{R} \right)$$

$$N = 70 \left(10 + \frac{200^2}{400} \right) = 70 \left(10 + \frac{40000}{400} \right) = 70(10+100) = 70 \times 110$$

Sol 18. Option (A)

Resolving forces along the radial direction towards the center:

$$T - mg \cos \theta = \frac{mv^2}{R} \Rightarrow T = \frac{mv^2}{R} + mg \cos \theta$$

Sol 19. Option (A)

At the top point, potential energy is $U_{\text{top}} = mg(2L) = 2mgL$.

Kinetic energy at top: $K_{\text{top}} = E - U_{\text{top}} = \frac{5}{2}mgL - 2mgL = \frac{1}{2}mgL$.

Since $K_{\text{top}} = \frac{1}{2}mv_{\text{top}}^2 \Rightarrow v_{\text{top}}^2 = gL$.

Tension at top:

$$T = \frac{mv_{\text{top}}^2}{L} - mg = \frac{m(gL)}{L} - mg = 0$$

Sol 20. Option (A)

With $u = \sqrt{6gL}$:

At lowest point: $T_L = \frac{m(6gL)}{L} + mg = 7mg$.

At highest point: $v_H^2 = 6gL - 4gL = 2gL \Rightarrow T_H = \frac{m(2gL)}{L} - mg = mg$.

Ratio:

$$\frac{T_L}{T_H} = \frac{7mg}{mg} = 7 : 1$$